

Supplementary Appendix: Per-Node RAF, Overhead, and PER Tables for Lightweight Cryptography for Secure Aggregation in Federated Learning for Network Anomaly Detection in Critical Infrastructures

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Abstract—This supplementary appendix accompanies the main paper and provides the complete per-node measurement, Resource Amplification Factor (RAF), Normalized Overhead (O), and Privacy-Efficiency Ratio (PER) tables for the five experimental cells: S1 (reference), S2_100ms, S2_250ms (high-latency network), and S3_70pct, S3_90pct (CPU-constrained edge). Each cell is averaged over 2 independent runs $\times$ 10 federated rounds. The RAF denominator is anchored to the S1 baseline across all cells, so cross-cell comparisons are commensurable. Tables are organized by scenario and split by client/server role to expose the role-asymmetric cost discussed in Section VII of the main paper.

Index Terms—Federated Learning, Secure Aggregation, Lightweight Cryptography, Edge Computing, Anomaly Detection

I. NOTATION AND CONVENTIONS

All tables follow the conventions established in Sections V and VI of the main paper. RAF is defined as $RAF_X = X_{\text{stack}}/X_{\text{baseline}}$ for $X \in \{L, CPU, RAM, BW\}$, with the S1 baseline used as the denominator regardless of the current scenario. Normalized overhead follows $O = \alpha RAF_L + \beta RAF_{CPU} + \gamma RAF_{BW} + \delta RAF_{RAM}$, with scenario-specific weights summarized in Table I. PER is defined as $PER = F_1/O$. Baseline rows have $O = 1$ and $PER = F_1$ by definition.

TABLE I: Scenario-Specific Weighting Profiles (Reproduced for Reference)

Cell	α (L)	β (CPU)	γ (BW)	δ (RAM)
S1	0.25	0.25	0.25	0.25
S2_100ms / S2_250ms	0.50	0.10	0.30	0.10
S3_70pct / S3_90pct	0.10	0.50	0.10	0.30

II. SCENARIO 1: REFERENCE CONDITION (S1)

Tables II–IV report the reference-condition measurements, RAF, and equally-weighted overhead/PER. These values establish the baseline anchor used across all subsequent cells.

TABLE II: S1 – Averaged Raw Measurements (mean $\pm$ std, 2 runs $\times$ 10 rounds)

Stack	Role	Latency (s)	CPU avg (%)	RAM avg (MB)	BW total (B)
Baseline	client	1.940 $\pm$ 0.170	0.238 $\pm$ 0.260	285.64 $\pm$ 6.67	1561.5 $\pm$ 5.1
Baseline	server	1.528 $\pm$ 0.112	0.826 $\pm$ 0.370	169.67 $\pm$ 0.05	3196.4 $\pm$ 7.5
Stack A	client	4.817 $\pm$ 6.286	0.688 $\pm$ 0.291	282.45 $\pm$ 10.9	3855.9 $\pm$ 213.5
Stack A	server	9.599 $\pm$ 11.02	0.462 $\pm$ 0.176	170.96 $\pm$ 0.64	8705.3 $\pm$ 229.6
Stack B	client	5.034 $\pm$ 6.234	0.844 $\pm$ 0.617	285.06 $\pm$ 7.21	3856.8 $\pm$ 213.3
Stack B	server	9.432 $\pm$ 10.28	0.514 $\pm$ 0.180	169.94 $\pm$ 0.23	8706.8 $\pm$ 228.5
Stack C	client	5.269 $\pm$ 5.105	3.293 $\pm$ 0.862	281.10 $\pm$ 6.16	3840.4 $\pm$ 213.8
Stack C	server	9.571 $\pm$ 8.637	2.188 $\pm$ 0.631	169.72 $\pm$ 0.07	8658.3 $\pm$ 230.3

III. SCENARIO 2: HIGH-LATENCY NETWORK

A. S2_100ms (100 ms injected one-way delay, 10 Mbps cap)

Tables V–VII report measurements under moderate latency stress. Stack A produces the lowest client-side overhead, while

TABLE III: S1 – Per-Node Resource Amplification Factor

Stack	Role	RAF_L	RAF_{CPU}	RAF_{RAM}	RAF_{BW}
Baseline	client	1.000	1.000	1.000	1.000
Baseline	server	1.000	1.000	1.000	1.000
Stack A	client	2.484	2.891	0.989	2.469
Stack A	server	6.281	0.560	1.008	2.724
Stack B	client	2.596	3.551	0.998	2.470
Stack B	server	6.172	0.622	1.002	2.724
Stack C	client	2.717	13.850	0.984	2.459
Stack C	server	6.263	2.649	1.000	2.709

TABLE IV: S1 – Per-Node Normalized Overhead and PER ($\alpha = \beta = \gamma = \delta = 0.25$)

Stack	Role	O	F_1	PER
Baseline	client	1.000	0.9788	0.979
Baseline	server	1.000	0.9788	0.979
Stack A	client	2.208	0.9783	0.443
Stack A	server	2.643	0.9783	0.370
Stack B	client	2.404	0.9783	0.407
Stack B	server	2.630	0.9783	0.372
Stack C	client	5.003	0.9783	0.196
Stack C	server	3.155	0.9783	0.310

server-side O rises across all stacks as the latency component is up-weighted to $\alpha = 0.50$.

TABLE V: S2_100ms – Averaged Raw Measurements

Stack	Role	Latency (s)	CPU avg (%)	RAM avg (MB)	BW total (B)
Stack A	client	4.533 $\pm$ 4.113	0.738 $\pm$ 0.357	284.84 $\pm$ 3.92	3856.1 $\pm$ 212.9
Stack A	server	8.403 $\pm$ 6.899	0.467 $\pm$ 0.200	166.83 $\pm$ 0.51	8705.2 $\pm$ 227.3
Stack B	client	4.650 $\pm$ 4.555	0.876 $\pm$ 0.689	285.29 $\pm$ 3.95	3856.6 $\pm$ 212.5
Stack B	server	8.597 $\pm$ 7.471	0.532 $\pm$ 0.194	166.61 $\pm$ 0.07	8708.1 $\pm$ 226.4
Stack C	client	5.301 $\pm$ 6.686	3.797 $\pm$ 0.909	285.27 $\pm$ 3.93	3839.6 $\pm$ 213.5
Stack C	server	9.766 $\pm$ 10.86	2.231 $\pm$ 0.657	166.64 $\pm$ 0.08	8656.9 $\pm$ 229.8

TABLE VI: S2_100ms – Per-Node RAF

Stack	Role	RAF_L	RAF_{CPU}	RAF_{RAM}	RAF_{BW}
Stack A	client	2.337	3.102	0.997	2.469
Stack A	server	5.498	0.566	0.983	2.723
Stack B	client	2.398	3.684	0.999	2.470
Stack B	server	5.626	0.644	0.982	2.724
Stack C	client	2.733	15.969	0.999	2.459
Stack C	server	6.390	2.701	0.982	2.708

TABLE VII: S2_100ms – Per-Node Overhead and PER ($\alpha = 0.50, \beta = 0.10, \gamma = 0.30, \delta = 0.10$)

Stack	Role	O	F_1	PER
Stack A	client	2.319	0.9783	0.422
Stack A	server	3.721	0.9783	0.263
Stack B	client	2.408	0.9783	0.406
Stack B	server	3.793	0.9783	0.258
Stack C	client	3.801	0.9783	0.257
Stack C	server	4.376	0.9783	0.224

B. S2_250ms (250 ms injected one-way delay, 10 Mbps cap)

Tables VIII–X report the per-node measurements, RAF, and overhead/PER under the higher latency-stress level. Latency-

weighted overhead remains stable across stacks, with Stack C continuing to lead the ranking due to its disproportionate RAF_{CPU} .

TABLE VIII: S2_250ms – Averaged Raw Measurements

Stack	Role	Latency (s)	CPU avg (%)	RAM avg (MB)	BW total (B)
Stack A	client	4.775 ± 4.919	0.794 ± 0.673	285.23 ± 3.90	3856.0 ± 212.6
Stack A	server	8.681 ± 7.721	0.507 ± 0.150	167.23 ± 0.29	8706.1 ± 226.8
Stack B	client	4.533 ± 4.127	0.898 ± 0.608	286.68 ± 3.13	3856.5 ± 213.6
Stack B	server	8.319 ± 6.734	0.543 ± 0.205	166.59 ± 0.07	8706.4 ± 229.4
Stack C	client	4.960 ± 4.811	3.730 ± 0.914	287.97 ± 0.50	3840.0 ± 213.2
Stack C	server	8.961 ± 7.750	2.291 ± 0.679	166.60 ± 0.07	8657.9 ± 228.7

TABLE IX: S2_250ms – Per-Node RAF

Stack	Role	RAF_L	RAF_{CPU}	RAF_{RAM}	RAF_{BW}
Stack A	client	2.462	3.340	0.999	2.469
Stack A	server	5.680	0.613	0.986	2.724
Stack B	client	2.337	3.777	1.004	2.470
Stack B	server	5.444	0.658	0.982	2.724
Stack C	client	2.558	15.687	1.008	2.459
Stack C	server	5.863	2.774	0.982	2.709

TABLE X: S2_250ms – Per-Node Overhead and PER ($\alpha = 0.50$, $\beta = 0.10$, $\gamma = 0.30$, $\delta = 0.10$)

Stack	Role	O	F_1	PER
Stack A	client	2.406	0.9783	0.407
Stack A	server	3.817	0.9783	0.256
Stack B	client	2.388	0.9783	0.410
Stack B	server	3.703	0.9783	0.264
Stack C	client	3.686	0.9783	0.265
Stack C	server	4.120	0.9783	0.238

IV. SCENARIO 3: CPU-CONSTRAINED MODE

A. S3_70pct (70% per-core background load)

Tables XI–XIII report measurements under moderate CPU contention. The CPU-weighted overhead profile sharpens the gap between stacks: Stack C’s client-side O rises to 8.89 while Stacks A and B remain near 2.2–2.4.

TABLE XI: S3_70pct – Averaged Raw Measurements

Stack	Role	Latency (s)	CPU avg (%)	RAM avg (MB)	BW total (B)
Stack A	client	4.552 ± 3.926	0.670 ± 0.265	284.65 ± 3.99	3856.7 ± 212.8
Stack A	server	8.295 ± 6.420	0.512 ± 0.164	166.71 ± 0.58	8706.8 ± 227.1
Stack B	client	4.418 ± 3.564	0.764 ± 0.226	283.94 ± 4.17	3856.6 ± 212.1
Stack B	server	8.017 ± 5.758	0.614 ± 0.235	166.61 ± 0.07	8708.2 ± 225.4
Stack C	client	4.771 ± 2.910	3.852 ± 0.855	285.77 ± 3.18	3841.0 ± 213.4
Stack C	server	8.159 ± 4.623	2.583 ± 0.650	166.20 ± 0.25	8658.8 ± 228.6

TABLE XII: S3_70pct – Per-Node RAF

Stack	Role	RAF_L	RAF_{CPU}	RAF_{RAM}	RAF_{BW}
Stack A	client	2.347	2.816	0.997	2.470
Stack A	server	5.428	0.619	0.983	2.724
Stack B	client	2.278	3.212	0.994	2.470
Stack B	server	5.246	0.744	0.982	2.724
Stack C	client	2.460	16.201	1.000	2.460
Stack C	server	5.339	3.127	0.980	2.709

B. S3_90pct (90% per-core background load)

Tables XIV–XVI report measurements under heavy CPU contention. Stack C’s RAF_{CPU} peaks at 16.70 $\times$ on the client side, producing the largest client-side O observed across all five experimental cells.

V. CROSS-CELL SUMMARY

Table XVII consolidates the client-side and server-side normalized overhead O across all five experimental cells. The

TABLE XIII: S3_70pct – Per-Node Overhead and PER ($\alpha = 0.10$, $\beta = 0.50$, $\gamma = 0.10$, $\delta = 0.30$)

Stack	Role	O	F_1	PER
Stack A	client	2.189	0.9783	0.447
Stack A	server	1.420	0.9783	0.689
Stack B	client	2.379	0.9783	0.411
Stack B	server	1.464	0.9783	0.668
Stack C	client	8.893	0.9783	0.110
Stack C	server	2.662	0.9783	0.367

TABLE XIV: S3_90pct – Averaged Raw Measurements

Stack	Role	Latency (s)	CPU avg (%)	RAM avg (MB)	BW total (B)
Stack A	client	4.543 ± 3.868	0.690 ± 0.209	282.51 ± 3.90	3856.6 ± 213.1
Stack A	server	8.239 ± 6.339	0.529 ± 0.198	167.22 ± 0.31	8707.1 ± 228.2
Stack B	client	4.618 ± 4.150	0.794 ± 0.241	282.58 ± 3.92	3856.0 ± 212.7
Stack B	server	8.295 ± 6.472	0.551 ± 0.186	166.60 ± 0.08	8705.7 ± 226.8
Stack C	client	4.899 ± 3.530	3.971 ± 0.836	283.90 ± 4.16	3840.6 ± 213.4
Stack C	server	8.630 ± 6.025	2.652 ± 0.724	166.61 ± 0.06	8660.2 ± 229.3

TABLE XV: S3_90pct – Per-Node RAF

Stack	Role	RAF_L	RAF_{CPU}	RAF_{RAM}	RAF_{BW}
Stack A	client	2.342	2.900	0.989	2.470
Stack A	server	5.391	0.641	0.986	2.724
Stack B	client	2.381	3.339	0.989	2.469
Stack B	server	5.428	0.667	0.982	2.724
Stack C	client	2.526	16.700	0.994	2.460
Stack C	server	5.647	3.211	0.982	2.709

TABLE XVI: S3_90pct – Per-Node Overhead and PER ($\alpha = 0.10$, $\beta = 0.50$, $\gamma = 0.10$, $\delta = 0.30$)

Stack	Role	O	F_1	PER
Stack A	client	2.228	0.9783	0.439
Stack A	server	1.428	0.9783	0.685
Stack B	client	2.451	0.9783	0.399
Stack B	server	1.443	0.9783	0.678
Stack C	client	9.147	0.9783	0.107
Stack C	server	2.736	0.9783	0.358

role asymmetry first reported in Section VI of the main paper persists across every cell: client-side O scales most aggressively for Stack C under CPU stress (reaching 9.15 at S3_90pct), while server-side O for Stacks A and B remains close to or below the baseline-aggregator cost under CPU stress ($O \approx 1.4$ for both at S3_70pct and S3_90pct), reflecting the $RAF_{CPU} < 1$ effect attributable to the PyCA C-speed implementation on the aggregator path.

TABLE XVII: Cross-Cell Summary of Normalized Overhead O (Client / Server)

Stack	S1	S2_100ms	S2_250ms	S3_70pct	S3_90pct
Stack A	2.21 / 2.64	2.32 / 3.72	2.41 / 3.82	2.19 / 1.42	2.23 / 1.43
Stack B	2.40 / 2.63	2.41 / 3.79	2.39 / 3.70	2.38 / 1.46	2.45 / 1.44
Stack C	5.00 / 3.16	3.80 / 4.38	3.69 / 4.12	8.89 / 2.66	9.15 / 2.74

REPRODUCIBILITY

The raw per-round measurement logs, post-processing scripts, and the configuration files used to generate every table in this appendix are available in the project repository: https://github.com/zackire/fl_sa_project.